

Exploring Causal Chain Identification: Comprehensive Insights from Text and Knowledge Graphs

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Abstract. During real-world reasoning, the logic path is generally not explicitly articulated. An appropriate causal chain can offer abundant informative details to depict a logical pathway, which is also beneficial in preventing ambiguity problems during text generation. However, most causal chains tend to lose their causal meaning after multiple hops, also this phenomenon occurs in other chains of relations. To discriminate the broken linkage in chain detection task, we introduce the **CK-CEVAE** model, **Chained domain Knowledge in Cause Effect Variational AutoEncoder**, which integrates knowledge into the representation of causal assumptions within chains, employing sequential probabilistic distributions for cause-effect estimation. Our model demonstrates an improvement of around 4% in F1-score over LLM-based and neural-based models in identifying causal chains originating from text. Furthermore, to investigate the semantic continuity of chains within established knowledge graphs, we curate a chain-structured dataset, highlighting both causal relations and multiple non-causal relations, i.e. *used for*, *synonym* and *similar to*, termed **ConceptNet-CC dataset**. We noticed that the longer the chains, the fewer instances of existence. However, contrary to our intuitions, models perform better at identifying longer chains than shorter ones in uni-directional relations like *causes* and *used for*.

Keywords: Causal Chain · VAE · Knowledge Graph

1 Introduction

In real-world reasoning, once the reasoning path becomes complex with multiple intermediate factors, the need for explicit expression becomes crucial. Causality serves as the primary element in reasoning and there is currently extensive exploration of textual causal pairs. However, one isolated causal pair is generally not sufficient to cover the entire logic of thinking. There is growing interest in forming causal chains, as a proper causal chain can furnish numerous informative details to flesh out a logical progression. However, in the majority of cases, the causal chain turns out to be meaningless from head term to tail term. For example, the two causal pairs <yoga, flexibility> and <flexibility, problem> are both meaningful respectively, but their causal chain <yoga, flexibility, problem> loses the semantic of causality. This phenomenon occurs not just for causality but also the non-causal relations.

This paper concentrates on detecting the successful causal chains and delving into the exploration of the semantic continuity in non-causal chains. Considering how cross-domain concepts

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disrupt causal chains, resulting in breaks, we propose the **CK-CEVAE** model, **Chained domain Knowledge in Cause Effect Variational AutoEncoder**. This model represents causal assumptions in chains, employing chained domain knowledge to identify broken linkages. Notably, it exemplifies the potential to improve Cause Effect VAE models (CEVAE)[10] for sequential cause-effect estimations on textual data, in contrast to the traditional focus of CEVAE on single-variable estimation in structured data. For those causal chains originated in text, CK-CEVAE demonstrates superior performance than other LLM-based models and neural-based models, i.e. GPT-4 [12], BiLSTM-CRF[5] and CVAE-SRNN[17], with the significant higher F1 and accuracy scores. While our model effectively detects causal chains from non-structural sources, it suffers from recall limitations for structural resources.

Furthermore, to support non-causal chains exploration, we generate a chain-structured dataset from ConceptNet[14], namely **ConceptNet-CC** dataset, and study the differences of information strength regarding two categories of relational chains: uni-directional and bi-directional relations. We noticed that the formation rate from pairs into chains is relatively low, varying between 0.53% and 2.16% across different relationships. The longer the chains, the fewer instances of existence. In the case study, we observed the uni-directional relations, e.g. *causes*, *used for*, exhibit stronger semantic continuity in message passing than bi-directional relations, e.g. *synonym*, *similar to*. However, contrary to our intuitions, models perform better at identifying longer chains than shorter ones in uni-directional relations like *causes* and *used for*.

This paper initiates by discussing relevant literature in Section 2. The methodology outlined in Section 3 comprises several pivotal elements, including the transformation of in-chain domain knowledge, the structure of sequential models, and the algorithms for chained prediction units. Following this, Section 4 introduces the utilized datasets in chains, with which we conduct an experimental analysis. Moreover, Section 5 presents a case study that investigates the semantic continuity for different relational chains. To conclude, the paper wraps up with an exhaustive discussion.

2 Related Work

Causal inference from text involves two main methodologies: measuring the causality of causal pairs by LLM-based methods and estimating cause effects of interest given a certain assumption among variables by neural-based methods.

Causal Inference by LLM-based methods The earliest approaches relied on finding the explicit causality in sentences. The explicit clues are a collection of causative verbs, causation adv./adj., causal links, and resultative constructions [1]. Getting rid of the crafty procedures at feature engineering, many language models like BERT [3] were examined and explored to detect causality directly from the labeled sentences or word pairs. To enhance causality detection from text, DTGNN [15] incorporated dependency relation features from a sentence to concatenate with the BERT encoder. Instead, ilab-FinCau [18] injected the causal knowledge injections into the encoders to significantly prevent the wrong identification of cause and effect parts. Despite advances in single causality detection, the ability to infer causal relationships when events are linked in a chain (like A causing B, and B causing C) is not well-developed, neither from text analysis nor from knowledge graph research. There is a study aiming to connect all causal tuples following the acquisition of causal pairs, leading to the creation of the causality knowledge graph FinCaKG [19]. However, validating their causal chains remains notably challenging.

In fact, given the widespread adoption of LLM-based models across diverse applications, framing causal inference as a classification problem offers a pragmatic approach: 1) training a sentence classifier using BERT by transforming concatenated causal chains into individual sentences; 2) embedding the linguistic information to DNN [8] for label prediction task on chains; 3) implementing the sequence labeling models like BiLSTM-CRF [5] for chained connection detection; 4). querying for generative models like GPT-4 [12] for question and answer tasks. However, to the best of our knowledge, the application of these prevailing methods for causal chain detection is limited, and there continues to be a deficiency in comparing those implementations within this field.

Causal Inference by neural-based methods Treatment effect estimation refers to a set of neural-based methods used to assess whether a treatment has a significant outcome based on the covariates of the observation data and to quantify the size of that effect [2]. A practical example, evaluating the treatment effects of two different medications for one disease, is the typical application of binary treatments. This conventional task has been extensively studied in observational studies. For example, TARNet[6] addresses the challenge of estimating treatment effects from observational data where treatments are not randomly assigned. While conditional VAE [13] is a more general framework that can be applied to various tasks where conditioning on additional information is necessary. From the observation data, covariates could be collected in different types, e.g. observed confounders and hidden confounders. The observed confounders are directly induced from the observation data, however, identifying all of the confounders is impossible in practice. We need further assumptions (hidden confounders) to estimate the treatment effects on the outcomes. CEVAE[10] was proposed to simultaneously estimate the unknown hidden space summarizing the hidden confounders and the cause effect. It is a probabilistic model that jointly modeling treatment assignment, potential outcomes, and observed outcomes, and aims to provide more accurate and robust estimates of causal effects compared to traditional methods. However, this model is primarily designed for binary outcomes and may not perform chained predictions. Comparatively, CVAE-SRNN [17] represents hidden confounders directly from causal pairs with conditional VAE and utilizes the structural neural networks for chained reasoning in causal chains. However, its precision suffers due to the strong assumption of treating all contextual information as pivot exogenous variables.

3 Methodology

For a causal chain of N components, e.g. A-B-C-D-E, we retrieve the embeddings of each component individually and combine the embeddings from the same causal pair by concatenating them in the final dimension, forming input variables X_n . With the presence of a causal chain instance, we depict this process to the left of Figure 1. Subsequently, our proposed CK-CEVAE model is introduced in the middle of the figure, followed by a demonstration of how the outcomes of chained CEVAEs are employed in the chain detection task.

3.1 In-chain Domain Knowledge

We posit that the comparison in domain knowledge between each component reflects the semantic alterations through chains. We would like to investigate the consistency of domain distribution inside each chain, denoting the components in chains by their sequence, e.g. head, 2nd, 3rd, etc..

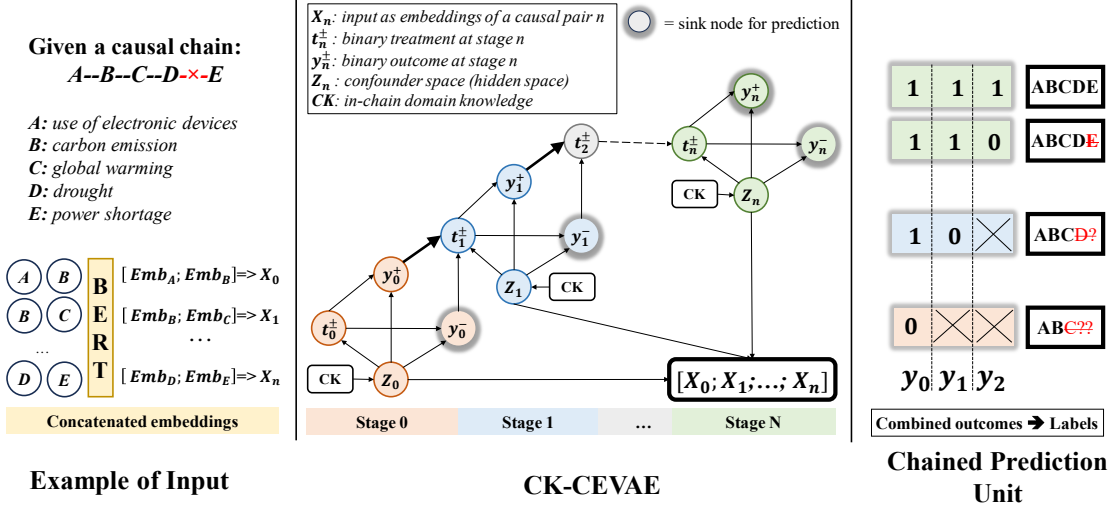


Fig. 1: The framework of CK-CEVAE for the chain detection task.

Due to the established semantic certainty of adjacent causal pairs in chains, we skip the first pair but start to compare the subsequent components, e.g. 3rd, 4th ..., to the head term one by one, termed as 'indirect causal pairs'. Analyzing these indirect causal pairs highlights the coherence of domain distribution within chains, namely **in-Chain domain Knowledge (CK)**. In this context, our investigation is restricted to 12 domains categorized according to Wikipedia classifications³.

For each indirect causal pair within a chain, we identify the top-k domains most closely associated with each component, with k set to 2 in our case. If both components of a pair share at least one common domain, we label this indirect causal pair as demonstrating positive domain consistency, unless negative. Thus, we pre-calculate the representation of in-chain domain knowledge for each input chain. The specifications for the domain detection model are discussed in Section 4.2. In the subsequent module, we will integrate the insights gained from analyzing domain distribution within chains into the proposed model.

3.2 CK-CEVAE

The vanilla Cause Effect VAE (CEVAE) [10] architecture is designed for structural data, aiming to simultaneously capture hidden confounders while estimating a solitary effect. To extend its applicability to text data, the preprocessing operation is essential to encode textual information into numerical representations. Based on the assumption that each causal pair within a causal chain represents the independent causality. As depicted in the left component of Figure 1, our approach proposes to pre-process a causal chain into distinct causal pairs, representing them as concatenated embeddings X_n , which serve as variables supporting the causality of each pair. The aggregation of these causal pair embeddings constructs the representation of a causal chain, illustrated as $[X_0; X_1; \dots; X_n]$, which serves as the input data for CK-CEVAE model, positioned at the center of Figure 1.

³ <https://en.wikipedia.org/wiki/Wikipedia:Contents/Categories>, access date: 2024/03/26

From the representation of causal chains, utilizing vanilla CEVAE enables the extraction of independent hidden confounders Z_n , independent treatments $t_n^\pm$, and independent outcomes $y_n^\pm$. However, this method only yields binary outcomes for each causal chain, which is unsuitable for our purposes of identifying a cohesive causal sequence within a lengthy chain. Moreover, it neglects the transitivity aspect in a chain, wherein the causal linkage from preceding pairs (AB) to subsequent pairs (BC) remains unaddressed. To incorporate this kind of chaining information effectively, we aim to establish connections between treatments and outcomes across causal pairs. Given that, within the same model, treatments should be independent of their potential outcomes ($(y^+, y^-) \perp t | X$), we adopt independent models for sequential estimation tasks while still preserving chained connections between these models. As depicted at the lower center of Figure 1, we name each estimation task of a causal pair as a *Stage* along the entire sequential tasks.

In *Stage 0*, the causal graph equipped with hidden confounder Z_0 and its treatment t_0 results in the binary outcomes y_0^+ and y_0^- . In the next stage, these outcomes y_0^+ and y_0^- are assigned as the observed treatments t_1 for each instance. Based on the same input embedding, the estimation procedure in *Stage 1* will generate the dedicated hidden confounder Z_1 and yield the outputs y_1^+ and y_1^- . In this manner, we connect n estimation tasks from *Stage 0* to *Stage n* together, by passing through the precedent outcomes to the following treatments, while keeping the hidden confounders unique and dedicated for local embeddings.

At startup, we initialize the treatment t_0 with all 1 by assuming the precedent outcomes are all positive. In the sequential stages, we bold the paths with positive outcomes or treatments to highlight the forwarding positive influences. For the termination of each path, we signify the sink of positive influences with the shaded nodes, i.e. $y_0^-, y_1^-, y_n^-, y_n^+$.

Generative network CEVAE assumes the observations factorize conditioned on the latent variables in the generative process, and uses an inference network that follows a factorization of the true posterior. In the generative procedure, we adhere to the core concept of the original CEVAE, but we enhance it by incorporating the generation of extra knowledge $\mathbf{g}_i$ conditioned on the hidden space. To simplify comparison, we use **bold** to highlight differences from the original formula.

Here the notations of stage number s_n are omitted, i.e. $z_i^{(s_n)}$ is simplified as z_i . For each stage, the model first generates the probability of hidden confounder z_i following normal distribution as $p(z_i)$. Conditioning on the generated confounder z_i , on the one hand, we calculate the distribution of extra knowledge $\mathbf{p}(\mathbf{g}_i | \mathbf{z}_i)$ and the distribution of input embeddings $p(x_i | z_i)$, on the other hand, we have the probability of binary treatment $p(t_i | z_i)$. Conditioning on the treatments and hidden confounder, $p(y_i | t_i, z_i)$ express the probabilities of the outcome y_i , where they both follow the Bernoulli distribution:

$$p(z_i) \sim \prod_{j=1}^{D_z} \mathcal{N}(z_{ij} | 0, 1); \quad \mathbf{p}(\mathbf{g}_i | \mathbf{z}_i) \sim \prod_{j=1}^{D_g} \mathbf{p}(\mathbf{g}_{ij} | \mathbf{z}_i) \quad (1)$$

$$p(x_i | z_i) \sim \prod_{j=1}^{D_x} p(x_{ij} | z_i); \quad p(t_i | z_i) \sim \text{Bern}(\sigma(f_1(z_i))) \quad (2)$$

$$p(y_i | t_i, z_i) \sim \text{Bern}(\pi = \hat{\pi}_i), \quad \hat{\pi}_i = \sigma(t_i f_2(z_i) + (1 - t_i) f_3(z_i)) \quad (3)$$

with $\mathbf{p}(\mathbf{g}_i | \mathbf{z}_i)$ and $p(x_i | z_i)$ being the appropriate probability distributions. Notice that i represents the number of instances, j is the covariate for different probabilities, i.e. in the case of $p(z_i)$, j ranges

to D_z the dimension of hidden space z ; for $\mathbf{p}(\mathbf{g}_i|\mathbf{z}_i)$, j ranges to $\mathbf{D}_g$ the number of features in extra knowledge $\mathbf{g}$; for $p(x_i|z_i)$, j ranges to D_x the dimension of x . For functions, $\sigma(\cdot)$ refers to the logistic function and $f_n(\cdot)$ for neural network parametrized by its parameters.

Inference network The hidden space z_i performs as the true posterior depends on x_i, t_i and y_i , the posterior approximation $q(z_i|x_i, t_i, y_i)$ calculated by x_i and y_i and conditioned on treatment t . This approximation architecture is similar to the classical inference network of CEVAE, here we omit the details. Instead, to preserve the chaining effect of sequential estimation, we focus on embedding the precedent information to the latter one through loss function formation, enhancing both training and inference processes with a focus on information transfer.

In each stage, we could form a single objective for both generative and inference networks, the variational lower bound is given by:

$$\begin{aligned} \mathcal{L}^{(s_n)} = \sum_{i=1}^C \mathbb{E}_{q(z_i^{(s_n)}|x_i, \mathbf{y}_i^{(s_{n-1})}, y_i^{(s_n)})} [\log p(\mathbf{g}_i | \mathbf{z}_i^{(s_n)}) + \log p(x_i, \mathbf{y}_i^{(s_{n-1})} | z_i^{(s_n)}) \\ + \log p(y_i | \mathbf{y}_i^{(s_{n-1})}, z_i^{(s_n)}) + \log p(z_i^{(s_n)}) - \log q(z_i^{(s_n)} | x_i, \mathbf{y}_i^{(s_{n-1})}, y_i^{(s_n)})] \end{aligned} \quad (4)$$

with C representing the instance count, (s_n) denoting objects served for computation in stage n . We highlight $\mathbf{y}_i^{(s_{n-1})}$, the output from stage $n-1$, which replaces $t_i^{s_n}$ in the original loss function to address the absence of model chaining information across stages.

Concerning out-of-sample predictions (unseen instances), theoretically, access to the treatment assignment t_i and its corresponding outcome y_i is essential for each stage before inferring the distribution over z_i . However, these values are unobservable except for the observed input x_i^* . Therefore, two auxiliary distributions are introduced to predict t'_i and y'_i for the masked instances during training:

$$q(t'_i | x_i^*) \sim \text{Bern}(\pi = \sigma(f_4(x_i^*))); \quad q(y'_i | x_i^*, t'_i) \sim \text{Bern}(\pi = \bar{\pi}_i) \quad (5)$$

In our notation, variables to predict are marked with $'$, while observed variables in the current stage are denoted with $*$. Due to the model's chaining effects, treatments can always obtained from the outcomes of the precedent stage. Therefore, for stage n , we adjust the variational lower bound to estimate the parameters of these two auxiliary distributions as:

$$\begin{aligned} \mathcal{L}_{predict}^{(s_n)} = \mathcal{L}^{(s_n)} + \sum_{i=1}^C [\log q(t_i'^{(s_n)} | x_i^*) + \log q(y_i'^{(s_n)} | x_i^*, t_i'^{(s_n)})] \\ = \mathcal{L}^{(s_n)} + \sum_{i=1}^C [\log q(\mathbf{y}_i^{*(s_{n-1})} | x_i^*) + \log q(y_i'^{(s_n)} | x_i^*, \mathbf{y}_i^{*(s_{n-1})})] \end{aligned} \quad (6)$$

with $y_i^{*(s_{n-1})}$ being observed outcomes from stage $n-1$ and $y_i'^{(s_n)}$ being outcomes to predict in stage n only.

3.3 Chained Prediction Unit

Variational Auto-Encoders (VAEs) [7], as a generative model, leverages probabilistic distributions, requiring numerous samples for a comprehensive assessment of its outcomes. In our case, the

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Function Sequential-Prediction ( $y_{i,m,r}^{s_n}, \theta, N, R, M$ )
  Input :  $y_{i,m,r}^{s_n}$  - An individual value of instance  $i$  in stage  $n$  as  $m$ -th sample at  $r$ -th randoms;
            $\theta$  - A threshold of variance tolerance among all sampling outcomes;
            $N, R, M$  - The number of models' stages, random states, and sampling results;
  Output:  $y_i$  - An overall outcome of instance  $i$ ;
  for  $n$  in  $N$  stages do
    for  $r$  in  $R$  random states do
      for  $m$  in  $M$  samples do
        if  $y_{i,m,r}^{s_n} > 0.5$  then  $y_{i,m,r}^{s_n} = 1$  else  $y_{i,m,r}^{s_n} = 0$ ; // Standardize initial values
      end
      if  $\sum_{m=0}^M y_{i,m,r}^{s_n} > M * \theta$  then  $y_{i,r}^{s_n} = 1$  else  $y_{i,r}^{s_n} = 0$ ; // Aggregate sampling results
      end
      if  $\sum_{r=0}^R y_{i,r}^{s_n} > R * 0.5$  then  $y_i^{s_n} = 1$  else  $y_i^{s_n} = 0$ ; // Obtain agreements among randoms
    end
  end
  return  $y_i = [y_i^{s_0}, y_i^{s_1} \dots y_i^{s_n}]$ 
end

```

Algorithm 1: The algorithm for deriving consolidated predictions from diverse sampling outcomes.

outcomes from CK-CEVAE are even complicated: we employ sequential models across different stages (i.e. N stages) with each model executed in distinct random states (i.e. R states) to eliminate stochastic influences. Due to the models' probabilistic nature, the results for each sampling operation (i.e. M samples) may exhibit considerable diversity even for the same input. To tackle this complex challenge, we introduce the sequential prediction algorithm to obtain a consolidated predictive output for each input. As outlined in Algorithm 1, except for the given outcomes $y_{i,m,r}^{s_n} \in \mathbb{R}^{N \times R \times M}$, our algorithm also takes into account the threshold θ of variance tolerance during sampling as well as the predetermined parameters N, R, M . These inputs lead to the prediction unit producing the overall outcomes y_i , presented as a sequence of binary labels for each instance i , which can serve as categorical labels for evaluation procedures. To facilitate understanding, we furnish an example showcasing the mapping from sequential binary to categorical labels, situated on the right side of Figure 1.

4 Experiments

4.1 Chains Acquisition

Relational chains, like causal chains, consist of sequentially connected word pairs with the same relation. Causal chains are a subset of them, connected by the relation 'causality'. Actually, it is not easy to acquire relational chains, because current knowledge graphs are designed to include and present all types of relations together. In this section, we talk about what kinds of relations are interesting and how to acquire the required relational chains from different resources.

Relations	Datasets	Examples of Relational Chains	Labels
causes	CCR-EN	[yoga, flexibility, problems, stress, marital problems]	Len2
		[excess estrogen, hormonal imbalance, acne, bacteria, gingivitis]	Len5
	ConceptNet-CC	[maintaining muscle strength, maintaining good health, living for long time, art objects, non believer]	Len3
		[taking final exams, passing class, graduating, getting job, needing to pay taxes]	Len4
used for		[stick eraser, pencil, erasing, self conscious, polynymous]	Len3
synonym		[hold, have, accept, bear, disinter]	Len4
similar to		[sentimental, tender, young, little, emotional]	Len5

Table 1: The instances of dataset **CCR-EN** and **ConceptNet-CC**.

Chains from text The chain-structured dataset **CCR-EN**⁴, proposed by Kai et al. [17], exclusively concentrates on causality from text in chains. Their causal chains, spanning 5 terms, draw from the corresponding causal pairs of referenced sentences, showcasing natural mentions reflective of real-world scenarios. Yet, there are no explicit tags for these terms to be associated with any Knowledge Graphs (KGs). The causality depicted within such causal chains is regarded as implicit knowledge lacking structured representation, posing a significant challenge for existing models.

For instance, Table 1 lists the relations and examples alongside their ground truth labels. Each chain comprises 5 terms in sequence, and the label reflects the extent to which the relation is successfully conveyed from the head term. ‘Len2’ denotes the relation extends to the 2nd term, e.g. ‘yoga’ causes ‘flexibility’ but does not continue to result in ‘problems’. While ‘Len5’ signifies the relation successfully extends to the 5th term, e.g. ‘excess estrogen’ causes ‘hormonal imbalance’, leading to ‘acne’, then ‘bacteria’, and finally resulting in ‘gingivitis’. Moreover, the dataset contains the original sentences where the causal pairs originated, serving as the textual context for those terms. It is important to note that only this dataset has context for causal chains, not the subsequent dataset.

Chains from Knowledge Graphs Obtaining relational chains is not straightforward, as current knowledge graphs are crafted to encompass a multitude of relations in unison. In this section, we explore the types of relations that pique interest and the method to extract the relational chains from existing knowledge bases.

We seek to understand how semantic information is conveyed through different kinds of relational chains. Our focus initiates with two relation categories: **bi-directional**, exemplified by *synonym*, *similar to*, and **uni-directional**, represented by *used for*, *causes*. We choose these relations because they connect terms with similar characteristics to those connected by ‘causes’ relations like the CCR-EN dataset, specifically focusing on **concepts**. Put simply, we avoid relations linking **name entity** terms. Consider, for example, "Barack Obama born in Honolulu", where both subject and object terms are identified as name entities. While its relation ‘born in’ is notably less compelling for investigating its chaining effects. Besides, the selected relations are typical examples of ‘uni-

⁴ <https://github.com/waste-wood/reco>, access date: 2024/03/26

Resources	Relations	Pairs	Len3	Len4	Len5	Chain Ratio (Len3/Pairs)
CCR-EN	uni causes	4,640	305	254	543	6.57%
	uni causes	33,481	682	384	264	2.04%
	used for	79,574	1,715	2,172	3,267	2.16%
ConceptNet-CC	bi synonym	400,609	2,117	1,433	1,443	0.53%
	similar to	60,560	551	468	350	0.91%

Table 2: The statistics of different relations in dataset **CCR-EN** and **ConceptNet-CC**. *Notes: ‘uni’ for ‘uni-directional’ category and ‘bi’ for ‘bi-directional’ category.*

directional’ or ‘bi-directional’ relations in commonsense knowledge graphs. To locate examples of the aforementioned relational chains, please refer to Table 1.

Though relational chains exist in knowledge graphs, most chains encounter difficulties in passing continuous information from head to tail. The difficulty is in choosing chains with reliable manners. In the extraction of positive relational chains, we adhere to this rule: a relational chain is deemed positive if its head and tail are directly connected by the same relation. ConceptNet [14] is a well-established knowledge base that incorporates crowd-sourced knowledge from sources like DBPedia [9], WordNet [11], Wikipedia⁵, and more. Utilizing ConceptNet-5.7 as the main resource, we form the relation chains from connected pairs and generate the dataset **ConceptNet-CC**. As shown in Table 2, we illustrate the quantity of its related pairs in the **Pairs** column. Additionally, we present an analysis of the statistics for positive relational chains identified at varying lengths 3, 4, and 5 alongside the **Chain Ratio** column, which indicates the likelihood of different relations forming a chain.

Furthermore, we provide the relevant statistics of the CCR-EN dataset in the same table. We noted that bi-directional pairs are generally plentiful, yet their chain ratios are consistently low. Regarding uni-directional relations, *used for* and *causes* exhibit relatively high chain ratios. Comparatively, uni-directional relations demonstrate enhanced chaining capabilities. Particularly for *causes* in CCR-EN, the chain ratio is approximately three times higher compared to the uni-directional relations from ConceptNet-CC. This is mainly because the CCR-EN [17] dataset was specifically created for discovering causality from text. Moreover, we observe a general decrease in most relations from short to long chains, except for *used for* and *synonym*. This deviation can be attributed to their long chains sharing a common short chain as a foundation in most cases, with *used for* being a notable example. This dataset **ConceptNet-CC** is available here⁶, along with a file indicating its 8:1:1 split for training, validation, and testing set.

4.2 Domain Detection Model

DeBERTa [4] stands out in capturing intricate semantic associations, making it particularly suitable for domain classification. It achieves an accuracy rate of around 90.6% on the MNLI-m

⁵ <https://www.wikipedia.org/>, access date: 2024/03/26

⁶ <https://drive.google.com/file/d/1msKbuACO7KUv-DlaekiBq4RbqJvBf5kg>, we intend to switch to a permanent link once accepted.

dataset [16] and holds a prominent position on the leaderboard⁷. We employ this model⁸ for zero-shot classification on 12 Wikipedia categories. After iterating through all indirect causal pairs within a sequence, we convert the aforementioned binary container into decimal digits, representing the domain distribution information for an entire chain.

4.3 Models Configurations

During training, in the presetting of sequential models structure of CK-CEVAE, we set the number of stages N to 3 considering the length of chains being 5, the number of sampling M for each operation is 100 and each model runs with 10 different random states R . For the technical parameters of each model, we list the value of each element: epoch to 500, latent dimension for confounder space to 50, hidden dimension to 200, number of neural layers to 2, learning rate to $5e-4$, and weight decay to $1e-4$. As for the comparison models, using some prevailing LLM-based and neural-based models straight out of the box isn't ideal for causal chain detection. Here, we outline the different pre-processions of each model to demonstrate how they can be effectively employed with the relational chains:

GPT-3.5/4 [12]: A LLM-based model using provided prompts to generate responses tailored to the given context or question. Employing few-shot prompting on both GPT-3.5-turbo and GPT-4, for each prompt, we randomly sample three chains (and their context if have) accompanied with gold labels from the training set of the same relation and ask for the label given a relational chain from the corresponding test set.

BERT [3]: A feedforward neural network on the pre-trained language models for the sequence classification task. We use it to categorize a sequence of five terms of a chain into a label denoting the different successful lengths of a chain. The training parameters like epochs are the same as CK-CEVAE but with an early stop.

Sequential DNN [8]: Following the sequential models structure of CK-CEVAE, instead, in each stage, we train an independent Deep Neural Network (i.e. two-layer neural network) to predict the binary outcomes. Without the requirements of a sequential prediction unit, we simply combine the binary predictions in each stage for categorical evaluation. For a better comparison, we apply with the same parameters as CK-CEVAE, e.g. epochs, learning rate, and hidden dimensions. The lack of causality linkage between different stages is the main difference to CK-CEVAE.

BiLSTM-CRF [5]: Bidirectional LSTM with a Condition Random Field layer is practically sufficient for sequence tagging tasks, it not only efficiently uses both past and future features of adjacent terms but also considers the tag information in a whole sequence. We transform our chain identification tasks into sequence tagging tasks by tagging terms with the BIO scheme. For instance, for each chain with length 5, we assign "BIOOO" for each term for a positive instance at Len2, and assign "BIIII" for a positive instance at Len5. In this way, each chain is tagged and each term in a chain will be predicted with a tag.

CVAE-SRNN [17]: This model uses conditional VAE to generate hidden space to represent the causality of each causal pair and propose the structural causal RNN to predict the causality of a subsequent term in a chain. Comparatively, CVAE-SRNN applied VAE independently for each causal pair but embedded the chaining connections in the stage of the proposed structural RNN.

⁷ <https://gluebenchmark.com/leaderboard/>, access date: 2024/03/26

⁸ <https://huggingface.co/MoritzLaurer/DeBERTa-v3-base-mnli-fever-anli>, access date: 2024/03/26

Models		CCR-EN				ConceptNet-CC			
		P	R	F1	Acc	P	R	F1	Acc
Prompting	GPT-3.5-turbo	19.64	26.59	21.78	25.71	30.27	31.25	26.59	31.25
	GPT-4	29.46	29.37	24.12	25.71	44.63	43.75	39.53	43.75
Fine-Tuning	BERT	71.66	81.17	76.12	66.53	90.38	97.92	94.00	91.67
	Sequential DNN	76.25	75.31	75.78	68.36	73.58	81.25	77.23	68.06
	BiLSTM-CRF	74.42	79.01	76.65	68.36	95.56	89.58	92.47	90.28
	CVAE-SRNN	66.3	94.75	78.01	64.90	70.17	83.33	76.19	65.28
	CK-CEVAE (ours)	74.16	88.58	80.73	72.21	96.67	80.56	87.88	85.19

Table 3: The overall results in the CCR-EN dataset and ConceptNet-CC dataset on *causes* relation.

Sources	Selected examples	Ground Truth	Prediction	
			CK-CEVAE	Other models
CCR-EN	[precursor tropical wave, thunderstorms, severe weather, flooding, extensive damage]	<u>Len3</u>	<u>Len3</u>	Len5
ConceptNet-CC	[visiting museum, seeing artifacts, learning about past, undefeatable, polygenesis]	<u>Len2</u>	Len3	<u>Len2</u>

Table 4: The selected examples showcase varying predictions between CK-CEVAE and comparative models.

4.4 Overall Analysis

We summarized the overall results of all comparative models in the CCR-EN dataset and ConceptNet-CC dataset on *causes* relations in Table 3. We observed that prompting models significantly lag behind fine-tuning models in both datasets, underscoring the limited capability of prevailing models like the GPT family in diagnosing causal chains. Particularly, they perform worse on text-originated causal chains (CCR-EN) than knowledge graph-originated ones (ConceptNet-CC).

For the CCR-EN dataset, our model can reach the highest F1 score and accuracy with a 2-4 percentage margin. Sequential DNN shows good precision but suffers a lot in recall, in contrast, CVAE-SRNN achieves the highest recall but the precision is the lowest. For a much more detailed analysis, in the comparison of sequential DNN and CK-CEVAE, the inclusion of chaining connections between sequential models increases a lot, showing the benefits of our coupling technique. As for BERT, it has been proven by its best chaining performance but CK-CEVAE generalizes a rather better-hidden space representation, resulting in the higher recall when the precision is close. Additionally, in the ConceptNet-CC dataset, our model reaches the highest precision among others but suffers from low recall as well. The effectiveness of sequence tagging models, such as BiLSTM-CRF and BERT, is evident in this dataset. It seems that we showed a good way to generalize this chaining identification task into a sequence tag recognition task.

Regarding the different performances of the two datasets, we believe that the CCR-EN dataset performs like open-world puzzles and the ConceptNet-CC dataset like close-world puzzles. Please

	P	R	F1	Acc
CK-CEVAE	74.16	88.58	80.73	72.21
- w/o DK	73.75	86.73	79.72	70.99
Embedding Replacement				
* Emb. CTX	73.63	87.04	79.77	70.99
* Emb. SENT	73.83	87.96	80.28	71.60

Table 5: The ablation study on the CCR-EN dataset⁹.

check the different predictions of selected examples in Table 4. The open-world instances (CCR-EN) are shown with the causality between concepts from natural phenomena to weather changes, while closed-world instances (ConceptNet-CC) are comprised of the multiple actions predefined in the knowledge graphs. In these instances, our model correctly identifies the causal link from ‘precursor tropical wave’ to ‘severe weather’ rather than disasters, but incorrectly generalizes the action "visiting museum" causing "learning about the past". For the structured causality in the ConceptNet-CC dataset, their knowledge is well captured from the large language model embeddings and the typical approaches are capable enough to identify them. But for the CCR-EN dataset, it turns out to be challenging. We believe the probabilistic models could represent well the causality assumptions from real-world text and use it for chaining identification.

4.5 Ablation Study

In order to show the effectiveness of knowledge injection in the proposed model, we employ the ablation study on different injection manners, including domain knowledge injection in hidden space, context knowledge fusion in embeddings, and sentence knowledge fusion in embeddings in Table 5.

We first remove the domain knowledge operation in the model, so as to delete the influence of domain knowledge in the hidden space distribution. The results of row *w/o DK* indicate a noticeable decline in all metrics. Keeping the absence of domain knowledge, we replace the original input embeddings with context embeddings and sentence embeddings. The former technique is to extract the term embedding from the mentions in its context, e.g. a causal sentence, the latter is to extract this sentence embedding directly. We observe a decline in performance for both experiments. This leads us to the conclusion that these embeddings introduce more uncertainty or chaos to our model in representing their causality assumptions. Furthermore, when we attempt to incorporate both domain knowledge injection and either sentence or context embedding fusion, we observe an even greater decline in performance. We infer that domain knowledge possesses abundant and unambiguous information for identifying chain effects, demonstrating the effectiveness of our approach in domain knowledge injection.

Relations	Size		Intruder Discrimination			Length Bias			
	Train	Test	SameRel	DiffRel	NoRel	Len2	Len3	Len4	Len5
causes	816	32	0.92	0.85	0.91	0.55	0.73	0.93	1.00
used for	5,396	636	0.95	0.96	0.93	0.86	0.86	0.91	0.95
synonym	3,216	32	0.61	0.69	0.65	0.59	0.41	0.22	0.22
similar to	952	80	0.62	0.65	0.69	0.20	0.32	0.18	0.26

Table 6: The F1-score of relational chains detection in ConceptNet-CC dataset.

5 Case Study: Understanding Semantic Continuity in Knowledge Graphs

Theoretically, there is broad consensus that, for one specific relation, the impact of semantic continuity during messaging passing decreases linearly with an increase in the number of hops. In reality, this consensus is not applicable since the information strength of different relations varies significantly. In this case study, besides causality, we investigate all other relations to understand the semantic continuity within current knowledge graphs, employing ConceptNet-CC as the underlying dataset.

Once we have positive chains, we propose to apply three different techniques to introduce the intruder components for those relational chains. Suppose we have a 3-length positive chain: A-B-C, we would like to replace the final word (i.e. C) with other terms (i.e. N) to have a negative chain A-B-N. The selection of replaced terms dominates whether this example is an easy negative or a difficult negative. We define three different techniques to replace the C term by considering its relation to the precedent term B. It is presented as the *intruder discrimination* column in Table 6. For example, if B and N hold the same relation as in the target chain, it is a difficult negative, tagged as **SameRel**. If B and N hold a relation but not the same as the target chain, tagged as **DiffRel**. If N has no relation to B but just is randomly selected, we consider it an easy negative and tag it as **NoRel**. In this way, we add up the intruder terms to the tail of positive examples until all positive chains reach length 5.

We analyze the semantic continuity of different relations by examining the challenges associated with their chain detection performance, utilizing the most typical model, BERT. Additionally, we deployed CK-CEVAE and regarded its results as additional information to affirm the overall performance of various relations. As discussed earlier, its performance, not outlined here, places it in a subordinate position compared to BERT concerning relations originating from the knowledge graph.

Generally, we summarized the performance of relational chain detection in Table 6. In the columns of *Intruder Discrimination* of Table 6, we observe that the uni-directional relations, e.g. *causes* and *used for* are less sensitive to three different intruders and turns to be easy to be detected. While bi-directional relations vary in a low F-1 score, conforming to our intuitions, the detection of chains is strongly influenced by the intruders with the same relations, but not others. As for intruders with different relations and no relations, their performance varies depending on different kinds of relations. Generally, we can conclude that uni-directional relations have higher

⁹ "w/o DK" notes for "without domain knowledge", "Emb. CTX" for term embedding with context information, and "Emb. SENT" for term embeddings replaced by its sentence embedding that a target term occurs.

information strength than bi-directional relations, owing to their steady and proficient performance, demonstrating reduced vulnerability to intruders. As for CK-CEVAE, we confirm this conclusion but with lower performance.

Except for information strength, we also examine the challenges associated with detecting chains of varying lengths. In the columns of *Length Bias* of Table 6, as the length grows, surprisingly the detection of uni-directional relation chains becomes more effortless, contrasting with the increased difficulty observed in bi-directional relation chains. We posit that the information strength of bi-directional chains diminishes significantly after a few hops, whereas for uni-directional chains, the information becomes more resilient and concrete. Therefore, we deduce that, within the context of the knowledge graph, uni-directional chains demonstrate stronger semantic continuity.

6 Discussion

To address the current limitations of ambiguity in text generation, exemplified by ChatGPT, the inclusion of causal or relational chains could be a vital recommendation before generation. If one seeks to request detailed statements or evidence of a causal pair, ChatGPT is often inclined to produce ambiguously all-purpose remarks. However, if the sequential keywords for the corresponding causal chain are provided before, the generated content becomes spot-on and situation-specific. We insist that extracting concrete relational or logical chains from the text, the primary manifestation of knowledge, reveals deeper insights than the constrained resources available in knowledge graphs. The introduction of a new algorithm is deemed necessary to effectively capture hidden causal chains in text, enabling their application to chain identification issues.

The strength of our model lies in its ability to identify the causal chains sourced from text, yet it falls behind the typical sequence tagging models when the causal chains are derived from knowledge graphs. In addition, the primary emphasis of this model lies in representing the hidden causal connections between variables and outcomes, but its versatility is constrained for other relations.

Nevertheless, it’s essential to highlight that in this study, our experiments practically reform diverse input formats to examine various comparative models on the specific chain detection task and report their performance across different datasets and relationships. We anticipate that this discussion will inspire other studies in chain detection, either originating from text or knowledge graphs.

7 Conclusion

We suggest that domain knowledge could be effective characteristics for distinguishing the continuity of chained relations. We propose CK-CEVAE to inject the in-chain domain knowledge for a better representation of the hidden causality in each stage and the effective sequential connections between different stages in chains. The results show that our model proves to be proficient in identifying causality originating from text but its effectiveness diminishes when faced with causality sourced from a knowledge graph. Furthermore, we offer the chain-structured ConceptNet-CC, designed for the relational chains detection task incorporating different levels of difficulty in handling intruder challenges. We conduct a thorough examination of the differences in semantic continuity for various relational chains of knowledge graphs. This investigation points to the conclusion that, in message passing, uni-directional relations have greater strength than their bi-directional counterparts. In the

future, we discern that the intriguing aspect centers on the selection or ranking of the most suitable chains from different candidates derived from the same relational pairs. The chains deemed most fitting are those appropriate for specific events or scenarios.

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